■■ Memory Error Encountered in Quantum Kernel SVM Training

Context

During the development of a quantum-classical comparative machine learning system for early medical diagnosis, a significant memory error was encountered while attempting to train a Quantum SVM (QSVM) model using the `QuantumKernel` from `qiskit-machine-learning`. The goal was to evaluate the feasibility and performance of a quantum kernel-based support vector machine in comparison with its classical counterpart on a real-world medical dataset.

Dataset Information

- Dataset: Cardiovascular Disease Dataset (sourced from Kaggle) - File: `cardio_train.csv` - Records: 70,000 samples - Features: 11 input features + 1 binary target - Preprocessing: Cleaned, scaled, and split using a custom `load_and_preprocess_data()` function from the `common.preprocessing` module.

System Details

- Python Version: 3.10.11 - Qiskit Version: 0.44.1 - Qiskit Machine Learning Version: 0.6.0 - Backend Used: Aer `statevector_simulator` - Feature Map: `ZZFeatureMap` with 11-dimensional input and 2 repetitions - Model: `SVC(kernel=quantum_kernel.evaluate)` from scikit-learn

Observed Error

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The error occurred during the kernel matrix computation step using
`quantum_kernel.evaluate(X_train, X_train)` internally within `SVC.fit()`. Full
traceback (truncated): ``` MemoryError: std::bad_alloc ``` This suggests that the
kernel evaluation attempted to allocate more memory than available on the system.
Given the size of the dataset (70,000 x 11), the kernel matrix alone would be
massive (70,000² = 4.9 billion entries), leading to exponential memory usage during
simulation.
```

Reflections

This issue was discovered while experimenting with quantum-enhanced models for real-world medical datasets. It highlights the current scalability challenges of quantum kernel methods on classical simulators like `statevector_simulator`. While quantum SVMs are promising, their practical application remains limited on large datasets without access to real quantum hardware or better-optimized simulators.